

KERRY DEAN JR.

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PROFESSIONAL SUMMARY

AI Engineer with 9+ years shipping production systems across federal, healthcare, and enterprise. Last two years on Large Language Model (LLM) products: Retrieval Augmented Generation (RAG), document understanding with GPT-4 and Claude, agentic systems with function calling and tool use, and clinical AI under FHIR / HIPAA. Currently shipping a Claude Agent SDK and n8n portfolio with custom Model Context Protocol (MCP) servers and skills. Strong across agentic, RAG, embeddings, vector search, fine-tuning, and evaluation.

EDUCATION

University of Iowa
Bachelor's, Computer Science

Iowa City, IA, USA
August 2012 - May 2016

SELECTED AI PROJECTS (INDEPENDENT, 2025 TO 2026)

Claude Agent SDK Production Skills

- Authored five production Claude Agent SDK skills covering productized agentic workflow generation, procedural KDP publishing pipelines, an Etsy API client with image generation, and browser automation for KDP. Each follows a deterministic scaffold plus LLM hybrid: code for correctness critical paths, prompt engineering for niche discovery and copy, with retry, fallback, and validation gates between stages. Open sourced template at github.com/whodeanie/claude-skill-template.

Multi Marketplace AI Product Suite

- Designed and operate an end to end content generation pipeline producing 50+ digital products across Gumroad, Etsy, and Amazon KDP: coloring books, puzzle books, niche information PDFs, and n8n workflow packs. Stack: Anthropic Claude API, OpenAI GPT-4, LangChain, Pinecone, vector embeddings, image generation, and programmatic listing creation. Storefront at kerryaiperson.gumroad.com.

n8n Agentic Workflow Library

- Built a library of agentic n8n workflow patterns covering function and tool calling, RAG retrieval, hybrid vector search, and multi step agent orchestration with first class error handling, retry policies, and integrations across Slack, Gmail, HubSpot, Airtable, Notion, OpenAI, and Anthropic. Each pack ships with per workflow setup docs, validation tests, and listing copy generated programmatically. Public showcase at github.com/whodeanie/n8n-agentic-workflows.

PROFESSIONAL EXPERIENCE

Traverse Technologies
Senior Technical Lead

Remote
June 2025 - November 2025

- Served as lead engineer on Dash, Traverse's corporate travel product for large enterprises to manage direct airline programs and onboard/offboard business travelers using their existing work credentials.
- Built and extended Dash features around user lifecycle, SSO, and directory sync using TypeScript/React on the frontend and C#/NET services on the backend, integrating with Okta and Azure AD.
- Designed and implemented backend services and APIs in C#/NET with SQL on AWS/Azure to handle traveler profiles, booking data, and policy rules, keeping Dash in sync with airline partners and internal corporate systems.
- Evolved CI/CD pipelines using GitHub Actions and Docker, adding automated tests, environment-specific configs, and gated deployments.
- Improved developer velocity by standardizing patterns, strengthening code reviews, and adding better logging and monitoring, making it easier to troubleshoot production issues and ship new Dash features safely.
- Architected and shipped the Choice Hotels integration, designing the partner API contract, building availability search and booking confirmation flows, handling rate mapping and loyalty program sync, and coordinating the phased rollout across enterprise accounts.

WellBe Senior Medical
Senior Software Engineer

Chicago, IL, USA
June 2023 - February 2025

- Developed the Luminate Web/Mobile Application, integrating advanced AI-driven automation with .Net Entity Framework, TypeScript, Python, Firebase, and AWS to enhance senior healthcare services.
- Built predictive analytics models enabling proactive patient care and streamlined provider workflows through intelligent scheduling and digital assessments.
- Built secure FHIR-compliant APIs exchanging patient data with health plan systems, pulling eligibility and claims history and pushing visit documentation back for care coordination.

- Managed a team of seven engineers, guiding agile processes and efficient DevOps pipelines.
- Created the clinical assessment module where providers complete standardized evaluations on tablets during home visits, with offline-first architecture using Firebase that syncs to AWS when connectivity returns.

FwdThink

Senior Generative AI Software Engineer

Remote

June 2024 - December 2024

- Built the document understanding pipeline processing incoming EEO complaints including PDFs, scanned forms, and photos of handwritten statements using AWS Rekognition for OCR, then passing extracted text through GPT-4 to identify complaint type, involved parties, alleged violations, and requested remedies.
- Developed the case routing system using LLM-based classification to assign complaints to the appropriate investigator queue based on jurisdiction, category, and workload balancing, replacing a manual triage process that was creating backlogs.
- Created the evidence analysis module processing supporting documents like emails, photos, and personnel records using Rekognition for image analysis and GPT for summarizing key facts and flagging potential corroborating evidence.
- Built the procurement research tool indexing contract databases, solicitation documents, and market reports into a vector store, letting acquisition officers ask natural language questions and get sourced answers.

Syngenta

Senior Software Engineer

Chicago, IL, USA

May 2021 - June 2023

- Developed scalable frontend and backend solutions for global agricultural analytics platforms using ReactJS, TypeScript, Python, AWS Lambda, GraphQL, and DynamoDB.
- Contributed core components to Syngenta's open-source React UI library including the map visualization widget, data grid with virtual scrolling, and chart components now used across the organization.
- Built the field monitoring dashboard displaying real-time satellite imagery, weather overlays, and sensor data on interactive maps where agronomists can zoom from regional views down to individual field boundaries and see current conditions.
- Enhanced real-time data processing capabilities, enabling farmers to make informed decisions to optimize crop yields and minimize risks.

Argonne National Laboratory

Software Developer Consultant

Chicago, IL, USA

November 2020 - April 2021

- Developed ETL pipelines in Python pulling emissions data from EPA databases, vehicle telemetry from fleet management systems, and electricity grid carbon intensity from EIA, transforming and loading into Neo4J for graph-based analysis.
- Created the comparative analysis dashboard visualizing total cost of ownership scenarios helping fleet managers build the business case for electrification.
- Collaborated closely with cross-functional research teams, aligning software solutions with broader environmental policy goals.
- Built the fleet electrification modeling tool letting researchers input fleet composition, routes, and usage patterns, then simulating emissions impact and infrastructure requirements of transitioning to electric vehicles.

Enablon

Software Engineer

Chicago, IL, USA

February 2018 - October 2020

- Developed and enhanced sustainability-focused web applications utilizing ReactJS, Redux, TypeScript, Node.js, C#, AWS, and SQL Server.
- Upgraded legacy client solutions to modern frameworks, significantly improving application performance, user experience, and maintainability.
- Managed complex data integrations and backend services, ensuring seamless interactions between frontend components and cloud-based resources.
- Collaborated with global teams to align technical implementations with strategic sustainability objectives.

Group O

Software Developer

Iowa City, IA, USA

October 2016 - January 2018

- Developed responsive web applications using React.js, Redux, Angular.js, Java, and Bootstrap, ensuring cross-browser and mobile compatibility.
- Created robust backend services with ASP.NET, C#, SQL Server, MongoDB, and PHP, enabling seamless data integration.
- Collaborated closely with stakeholders to translate business requirements into effective software solutions.
- Improved application performance and user experience through targeted enhancements and optimizations.

SKILLS

Skills: AI Engineering, Generative AI, Large Language Models (LLMs), Anthropic Claude API, Claude Agent SDK, OpenAI GPT-4, LangChain, LlamaIndex, Pinecone, Weaviate, FAISS, Retrieval Augmented Generation (RAG), Hybrid retrieval, Vector search, Embeddings, Reranking, Prompt engineering, Function calling, Tool use, Agentic workflows, Fine-tuning, Evaluation harnesses (LangSmith, Braintrust, Phoenix), Model Context Protocol (MCP),

PyTorch, Hugging Face, MLflow, Databricks, TypeScript, React.js, Next.js, Python, AWS (Lambda, Bedrock, Rekognition, S3), Microsoft Azure, Google Cloud Platform, Docker, CI/CD, GitHub Actions, CircleCI, Firebase, Vercel, Postgres, DynamoDB, BigQuery, Redis, Neo4j, C#, Java, SQL, GraphQL, REST APIs, FHIR, ASP.NET, Spring, Vue.js, Node.js, .NET, Entity Framework, JavaScript, C/C++, Kotlin, Flutter, Bootstrap, Figma, Git, Zapier, Agile.